INSERTION GUIDE

WHERE TO INSERT THIS SECTION:

Insert this entire section AFTER the section that introduces the physical

kiosk / self-service terminal hardware and BEFORE any section that discusses

ASR (Automatic Speech Recognition), NLP, or intent classification.

Rationale: Voice Capture is a preprocessing layer — it feeds clean audio

INTO the ASR engine. It must appear in the document BEFORE ASR is discussed

so the pipeline reads left-to-right: hardware → voice capture → ASR → logic.

HOW TO NUMBER IT:

Replace "X" in all heading numbers below with the next sequential number

in your report (e.g., if the preceding section is "4. Self-Service Kiosk

Architecture", this becomes "5. Voice Capture, Isolation & Enhancement System"

and its subsections become 5.1, 5.2, 5.3 ... 5.7).

WHAT TO DO AFTER INSERTING:

Increment all subsequent section numbers by 1.

No other changes to the original report are needed.

DELETE EVERYTHING ABOVE THIS LINE BEFORE INSERTING INTO THE REPORT

X. Voice Capture, Isolation & Enhancement System

X.1 Problem Context: Bank Branch Acoustic Environment

Physical bank branches are acoustically challenging environments for reliable automated speech interaction. Typical noise sources include multiple parallel conversations, queue chatter, teller interactions, printer and keyboard noise, HVAC/AC hum, and occasional impulsive sounds such as coin handling or door slams. These sources create a highly non-stationary, multi-speaker sound field around the self-service kiosk.

In addition, kiosk speakers (for voice bot playback), hard reflective surfaces (glass, tiles, counters), and relatively long reverberation times cause echoes and late reflections that smear the speech spectrum and degrade automatic speech recognition (ASR) performance, especially for low-SNR customers standing farther from the microphone. Reliable voice isolation in this context is critical to ensure that the downstream ASR, Deterministic Logic Bridge, and

transaction logic operate on clean, user-intent-focused audio rather than a mixture of customers and background conversations.

X.2 Selected Voice Processing Approach

To meet real-world banking constraints (low latency, robustness, cost efficiency, and privacy-preserving edge deployment), the platform adopts a focused, practical voice processing chain instead of a collection of heavy, research-grade separation models.

The selected approach combines:

- A small microphone array embedded in the kiosk
- Deterministic beamforming for spatial (direction-based) filtering
- Lightweight neural noise suppression for residual noise and interference removal
- Voice Activity Detection (VAD) to gate downstream processing and reduce load

This combination is chosen because:

- **Low latency:** Classical beamforming and VAD operate on short frames (10–20 ms), and lightweight AI denoisers such as RNNoise or DeepFilterNet-class models add only tens of milliseconds of algorithmic delay, keeping the end-to-end system comfortably under the 300 ms interaction latency target.
- **Real-time feasibility:** All components can run in real time on edge devices such as Raspberry Pi or NVIDIA Jetson modules without requiring a datacenter GPU; compute and memory footprints are bounded and predictable.
- **Robustness to bank conditions:** Spatial filtering from the array attenuates off-axis speakers and crowd noise, while neural suppression handles non-stationary background sounds and residual chatter that pass through the beam.
- **Edge deployability and privacy:** All multichannel raw audio remains on the kiosk. Only a single enhanced mono stream or ASR text leaves the device, maintaining compatibility with the existing Privacy Proxy, Edge Intelligence layer, and trust boundaries.

X.3 End-to-End Voice Processing Pipeline

The branch kiosk uses the following standardized voice pipeline before ASR and intent understanding:

Mic Array → Beamforming → Noise Reduction → VAD → ASR

1. **Microphone Array Capture**
   - 2–6 matched digital MEMS microphones are embedded around the kiosk display (e.g., in a linear top-edge or circular/"ring" geometry around the camera).
   - The array samples multichannel audio at 16 kHz or 24 kHz, synchronized via a shared I^2S/PDM interface and clock, providing spatial information about the sound field around the user.
2. **Beamforming (Spatial Filtering)**
   - A fixed or slowly adaptive beamformer is designed with its main lobe pointing toward the expected customer position in front of the kiosk.
   - Delay-and-sum or MVDR-style beamforming is applied to the multichannel input, producing a single "beamformed" signal where speech from the user is emphasized while sounds from other directions (side conversations, back-of-branch noise) are attenuated.
   - This spatial pre-filtering provides several dB of SNR improvement before any neural processing runs, reducing the load on subsequent stages.
3. **Noise Reduction (Neural Suppression)**
   - The beamformed signal is fed into a lightweight AI noise suppression module such as RNNoise or a compact DeepFilterNet-class model running on the edge device.
   - The denoiser operates frame-by-frame, estimating a spectral mask or time-domain correction that suppresses residual background speech, HVAC noise, keyboard clicks, and reverberation tails, while preserving the customer's speech characteristics for accurate ASR.
   - Model size and lookahead are chosen such that CPU utilization remains within the kiosk budget and algorithmic delay remains small (on the order of 20–60 ms).

4. **Voice Activity Detection (VAD)**
    - A VAD module analyzes the enhanced stream to determine when the user is actively speaking.
    - The VAD output is used to:
        - Gate audio frames sent to ASR, so silence and non-speech background noise are not transmitted or transcribed.
        - Control turn-taking logic in the dialogue manager (e.g., only respond after user speech segments end, avoid barge-in confusion).
        - Reduce compute and bandwidth, since only frames marked as "speech" need high-fidelity processing or uplink to cloud ASR when used.
5. **ASR and Downstream Logic**
    - Only the VAD-filtered, beamformed, denoised mono audio stream is fed into the ASR engine, ensuring that recognition operates on the cleanest possible representation of the customer's voice.
    - The ASR output (transcript and timing information) is then passed to the Deterministic Logic Bridge and higher-level AI components, preserving the existing architecture for intent classification, policy enforcement, and transaction orchestration.

X.4 Hardware Integration

The voice processing chain is designed to integrate into the kiosk hardware with minimal disruption:

- **Microphones:**
    - Use 2–6 low-noise omnidirectional MEMS microphones with matched sensitivity and frequency response.
    - Place them around the display area at known positions, enabling a fixed geometry beamformer tuned to the "typical customer position" at normal standing distance.
- **Edge Compute Device:**
    - Run beamforming, noise suppression, and VAD on the existing edge compute module (e.g., Raspberry Pi 4/5 or NVIDIA Jetson Nano/Orin Nano), alongside the Edge Intelligence components that already handle local decisioning and communication with the cloud back-end.
    - Ensure the audio processing stack is implemented in a high-performance language (C/C++ for the beamformer and RNNoise/DeepFilterNet runtime) with bindings to the application layer that manages ASR and interaction flows.
- **System Integration:**
    - The voice processing stack sits logically below or alongside the Edge Intelligence layer, exposing a "clean audio" interface to the rest of the system.
    - It does not alter or bypass the Privacy Proxy or any existing encryption, logging, or policy enforcement components; it simply improves the signal quality before those systems see the data.

X.5 Handling Multi-Speaker Scenarios

While the selected approach cannot perfectly separate fully overlapping speakers in all conditions, it significantly reduces the impact of other voices through a combination of spatial filtering, UX constraints, and dialogue design:

- **Directional Filtering via Beamforming**
    - The beamformer is oriented toward the active kiosk user, attenuating conversations occurring at different angles (e.g., people in the queue behind or to the side).
    - This spatial selectivity means that, in most practical situations, the loudest and clearest signal in the beam will be the user directly in front of the kiosk.
- **User Positioning and Physical Design**
    - The kiosk can subtly guide users to stand in an optimal zone (e.g., a floor marking or on-screen "Stand here and speak" indicator aligned with the beam).

    - Optional semi-enclosures or acoustic panels around the screen area further reduce lateral and rear noise pickup.
- **UX and Turn-Taking Control**
    - The conversational flow encourages the customer to speak only when prompted (e.g., "Please state your request after the beep"), reducing overlap with other environmental speech.
    - The VAD ensures that ASR is engaged primarily when the user is expected to speak, and background chatter during system prompts is ignored as much as possible.

In edge cases where multiple people speak simultaneously very close to the kiosk, the system may still capture mixed speech. In such scenarios, the Deterministic Logic Bridge and higher-level logic can detect low ASR confidence or inconsistent intents and request clarification from the user, rather than proceeding with uncertain transactions.

X.6 Real-Time Constraints and Latency

The voice processing system is explicitly designed to meet the overall interaction latency target of less than 300 ms from user speech to system response:

- **Frame Sizes and Lookahead**
    - Beamforming and VAD operate on short audio frames (typically 10–20 ms) with negligible additional latency.
    - The chosen lightweight noise suppression model is configured with minimal lookahead and buffering, keeping its contribution to latency on the order of tens of milliseconds.
- **Edge Execution**
    - Running all capture, beamforming, enhancement, and VAD on the kiosk edge device eliminates round-trip network latency for preprocessing and ensures deterministic performance even under variable network conditions.
    - Only the final enhanced mono stream (for cloud ASR) or recognized text (for on-device ASR) traverses the network, fitting within the existing latency budget of the AI and transaction layers.
- **Overall Budget Alignment**
    - With careful implementation, the voice front-end introduces only a small fraction of the total response time, leaving the majority of the latency budget to ASR, Deterministic Logic Bridge logic, back-end validation, and response generation.

X.7 Failure Modes and Mitigations

The voice isolation system includes clear handling strategies for common failure modes to maintain safety and user trust.

1. **Multiple People Speaking Near the Kiosk**
    - **Problem:** Two or more speakers stand very close to the kiosk and speak at the same time, causing overlapping speech even within the main beam.
    - **Mitigation:**
        - The system monitors ASR confidence scores and intent consistency; low confidence or conflicting hypotheses trigger polite clarification prompts (e.g., "Sorry, I could not clearly understand. Please speak one at a time and repeat your request.").
        - On-screen guidance and audio prompts instruct users that only one person should speak at a time when using the kiosk.
        - For persistent issues, the system can route the interaction to a human agent or suggest using a quieter area or another channel.
2. **Very High Background Noise or Acoustic Shock Events**
    - **Problem:** Sudden loud noises (e.g., dropped metal objects, alarms) or extended high ambient noise periods (e.g., peak hours) can temporarily overwhelm the beamformer and denoiser, reducing intelligibility.
    - **Mitigation:**

    - The VAD and signal-level monitors can detect severe SNR degradation or clipping and temporarily suspend ASR, displaying a message asking the user to wait a moment or move closer to the kiosk.
    - Automatic gain control (AGC) and dynamic thresholding prevent the system from reacting to extreme outliers as valid speech segments.
    - Branch operations can be configured to receive telemetry on repeated noise issues and adjust kiosk placement or local acoustics over time.

3. **Microphone or Channel Failure**
    - **Problem:** One or more microphones in the array fail or drift significantly in gain or frequency response, degrading the beam pattern and voice quality.
    - **Mitigation:**
        - The kiosk periodically runs a self-test or monitors per-channel noise floors and coherence to detect failing or severely imbalanced channels.
        - Beamformer logic can automatically fall back to a reduced-microphone configuration (e.g., using only the remaining healthy channels) with a slightly wider beam but maintained service.
        - Maintenance alerts are raised through the existing monitoring and operations channels so hardware can be serviced without silently degrading user experience.

Across all failure modes, the architecture ensures that the voice front-end never bypasses the Deterministic Logic Bridge, Privacy Proxy, or transaction safeguards. Instead, it either supplies high-confidence, clean input or deliberately degrades into a "safe failure" mode where it requests clarification, routes the interaction to a human, or temporarily suspends sensitive operations.